

NATIONAL INSTITUTE OF TECHNOLOGY ROURKELA
END - SEM EXAMINATION SUMMER, 2025
SESSION: 2024 – 2025 (Spring)
Guided-Self

Subject code: CS6510
No. of pages: 2

Subject Name: Deep Learning
Full Marks: 50

Dept. Code: CS
Duration: 3 Hrs

Answer ALL questions. Unnecessary and extra writing will attract negative marks.

Q. No.	Questions	Marks	CO	BL
1. (a)	What is an autoencoder? How does a denoising autoencoder differ from a vanilla autoencoder? Explain its application in noise-robust feature learning.	5+5	CO1	BL1
(b)	Given a query $Q = 1,2$, keys $K = [1,0, 0,2]$, and values $V = [1,1, 3,1]$. Use scaled dot-product attention (with $d_k=2$) to compute: - Dot product scores - Scaled scores - Softmax attention weights - Final output			
2. (a)	Explain the architecture of a Gated Recurrent Unit (GRU) and how it mitigates the vanishing gradient problem. Differentiate between GRU and a simple RNN with respect to memory retention.	5+5	CO2	BL2
(b)	(i) A simple RNN uses tanh activation. Given: Input $x_t = [0.2, -0.4]$, Previous hidden state $h_{t-1} = [0.3, 0.5]$ Input-to-hidden matrix $W = [0.1, -0.2, 0.4, 0.3]$, Hidden-to-hidden matrix $U = [0.3, 0.1, -0.2, 0.2]$, Bias vector $b = [0, 0]$. Calculate h_t .			
3. (a)	Compare AlexNet and GoogLeNet architectures in terms of layer design, parameter efficiency, and feature extraction strategy. Why are Inception modules used in GoogLeNet?	5+5	CO3	BL3
(b)	An input image of size $64 \times 64 \times 3$ passes through the following layers: - Conv layer: 32 filters of size 5×5, stride=1, padding=2 - Max pooling: 2×2 window, stride=2 - Another conv layer: 64 filters of size 3×3, stride=1, padding=1 Calculate the final feature map dimensions after each layer.			

4.	(a)	Explain the Encoder–Decoder architecture used in sequence-to-sequence learning. How does this model handle variable-length input and output? Provide an example of its application in machine translation.	5+5	CO2	BL2
	(b)	What role does positional encoding play in Transformer models? Why can't self-attention alone handle the sequence order? Illustrate with an example.			
5.	(a)	Discuss the architecture of a typical GAN. What are the limitations of a vanilla GAN, and how do DCGAN and CycleGAN overcome them?	5+5		
	(b)	Explain the architecture of U-Net for biomedical image segmentation. Additionally, describe how the Bellman Equation is used in Q-learning within the reinforcement learning paradigm.		CO3	BL3

CO: Course Outcome and BL: Bloom's Level

***** ALL THE BEST *****